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Big Data Analytics

BDA101 | Suggested sequence · Semester 4 | Student-authored study guide

How to use these notes

Distributed storage and computation, partitioning, batch/stream processing, and scalable analytics trade-offs. Work through the units in order, reproduce each example, then answer the review prompts from memory. This is supplemental practice material, not an official MBU syllabus or assessment. Course sequencing is suggested and should be checked against your current approved syllabus.

MODULE 01 · Scale, partitions, and data formats

Large workloads are divided across machines. Partitioning enables parallel work but can cause skew; columnar formats help analytical scans. Track schema, provenance, retention, and access control.

Worked example

```
Partition event data by date for time-bounded scans, while avoiding an excessive number of tiny files.
```

Check your understanding

- What is data skew?
- Why use a columnar file for scans?

MODULE 02 · Distributed processing

Map-style operations transform partitions and reduce-style operations aggregate results. Shuffles move data across the network and are expensive. Minimize unnecessary shuffles and plan for retries.

Worked example

```
Compute regional totals by mapping each event to (region, value), then reducing by region.
```

Check your understanding

- What is a shuffle?
- Why must distributed tasks tolerate retries?

MODULE 03 · Batch, streams, and governance

Batch processing handles bounded data; stream processing handles ongoing events with explicit time and late-data semantics. Monitor freshness, correctness, cost, privacy, and deletion requirements.

Worked example

```
A daily report can use batch processing; fraud alerts may require event-time windows and late-arrival handling.
```

Check your understanding

- What is event time?
- Name a governance concern for event streams.

Revision checklist

- %¡ Can I explain the main idea without copying the notes?
- %¡ Can I reproduce the example and state its assumptions?
- %¡ Can I test an edge case and describe what should happen?
- %¡ Have I recorded a limitation and checked a primary reference?

Further reading

Use the official documentation and open references linked in the portal. Check the publication/version and license at the source before redistribution. Cite the source and access date in coursework.